THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 11	Replace x by its value and calculate.
2	**(d)** 11	Replace y by its value and subtract.
3	**(b)** $n + 5$	Translate the words into a letter-number expression.
4	**(a)** 3	Terms are separated by + or - signs.
5	**(c)** 7	The coefficient is the number multiplying the letter.
6	**(a)** $3x$ and $9x$	Like terms have the same letter part.
7	**(a)** 12	Add the two values.
8	**(d)** 20	Multiply the coefficient by the value of x.
9	**(d)** 16	Replace x by its value and calculate.
10	**(a)** 5	Replace y by its value and subtract.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substitute: $2(3) + 5$ (1); $= 11$ (1).	1+1
12	Identify the operations (1); Expression: $4n + 7$ (1).	1+1
13	Group the like terms (1); Sum $= 5x + 5$ (1).	1+1
14	Substitute: $4(6) + 1$ (1); $= 25$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Substitute: $2(4) + 3(5)$ (1); $= 8 + 15$ (1); $= 23$ (1).	1+1+1
16	All three terms are like terms (1); Coefficients: $7 + 4 - 3 = 8$ (1); Answer: $8x$ (1).	1+1+1
17	Cost $= 5p$ (1); At $p = 12$: 5×12 (1); Cost $=$ ₹60 (1).	1+1+1
	OR: Substitute: $3(6) + 2(7)$ (1); $= 18 + 14$ (1); $= 32$ (1).	1+1+1
18	All three terms are like terms (1); Coefficients: $9 + 6 - 5 = 10$ (1); Answer: $10x$ (1).	1+1+1
	OR: Cost $= 8p$ (1); At $p = 15$: 8×15 (1); Cost $=$ ₹120 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sister's age $= x - 3$ (1); Sum $= x + (x - 3) = 2x - 3$ (1); At $x = 15$: sister $= 12$ years (1); Sum $= 27$ years (2).	1+1+1+2
	OR: Perimeter $= 2(l + 4)$ (1); Area $= 4l$ (1); At $l = 9$: perimeter $= 26$ cm (1); Area $= 36$ cm^2 (2).	1+1+1+2
20	Sister's age $= x - 5$ (1); Sum $= x + (x - 5) = 2x - 5$ (1); At $x = 18$: sister $= 13$ years (1); Sum $= 31$ years (2).	1+1+1+2
	OR: Perimeter $= 2(l + 5)$ (1); Area $= 5l$ (1); At $l = 12$: perimeter $= 34$ cm (1); Area $= 60$ cm^2 (2).	1+1+1+2

Mistakes to avoid:

- Adding unlike terms: $4x + 3y = 7xy$ is wrong.
- Writing $3a + 2a = 5a^2$; the letter part does not change when like terms are added.
- Writing $3n$ as '3 plus n'; it means 3 times n.
- Not changing all signs when subtracting an expression in brackets: $(5x + 3) - (2x - 4) \neq 3x - 1$.
- Confusing 'n less than a' $(a - n)$ with $n - a$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $3m$	Translate the words into a letter-number expression.
2	**(d)** 2	Terms are separated by + or - signs.
3	**(b)** 3	The coefficient is the number multiplying the letter.
4	**(d)** $4y$ and $-y$	Like terms have the same letter part.
5	**(b)** 17	Add the two values.
6	**(a)** 27	Multiply the coefficient by the value of x.
7	**(b)** 17	Replace x by its value and calculate.
8	**(a)** 13	Replace y by its value and subtract.
9	**(c)** $p - 7$	Translate the words into a letter-number expression.
10	**(c)** 4	Terms are separated by + or - signs.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Identify the operations (1); Expression: $2m - 3$ (1).	1+1
12	Group the like terms (1); Sum $= 9x + 8$ (1).	1+1
13	Substitute: $3(5) + 7$ (1); $= 22$ (1).	1+1
14	Identify the operations (1); Expression: $5(a + 2)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Substitute: $5(2) + 4(3)$ (1); $= 10 + 12$ (1); $= 22$ (1).	1+1+1
16	All three terms are like terms (1); Coefficients: $12 + 5 - 8 = 9$ (1); Answer: $9x$ (1).	1+1+1
17	Cost $= 6p$ (1); At $p = 20$: 6×20 (1); Cost $=$ ₹120 (1).	1+1+1
	OR: Substitute: $4(8) + 5(9)$ (1); $= 32 + 45$ (1); $= 77$ (1).	1+1+1
18	All three terms are like terms (1); Coefficients: $15 + 3 - 9 = 9$ (1); Answer: $9x$ (1).	1+1+1
	OR: Cost $= 9p$ (1); At $p = 11$: 9×11 (1); Cost $=$ ₹99 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sister's age $= x - 4$ (1); Sum $= x + (x - 4) = 2x - 4$ (1); At $x = 20$: sister $= 16$ years (1); Sum $= 36$ years (2).	1+1+1+2
	OR: Perimeter $= 2(l + 3)$ (1); Area $= 3l$ (1); At $l = 10$: perimeter $= 26$ cm (1); Area $= 30$ cm^2 (2).	1+1+1+2
20	Sister's age $= x - 2$ (1); Sum $= x + (x - 2) = 2x - 2$ (1); At $x = 12$: sister $= 10$ years (1); Sum $= 22$ years (2).	1+1+1+2
	OR: Perimeter $= 2(l + 6)$ (1); Area $= 6l$ (1); At $l = 8$: perimeter $= 28$ cm (1); Area $= 48$ cm^2 (2).	1+1+1+2

Mistakes to avoid:

- Adding unlike terms: $4x + 3y = 7xy$ is wrong.
- Writing $3a + 2a = 5a^2$; the letter part does not change when like terms are added.
- Writing $3n$ as '3 plus n'; it means 3 times n.
- Not changing all signs when subtracting an expression in brackets: $(5x + 3) - (2x - 4) \neq 3x - 1$.
- Confusing 'n less than a' $(a - n)$ with $n - a$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 9	The coefficient is the number multiplying the letter.
2	**(d)** $7ab$ and $2ab$	Like terms have the same letter part.
3	**(b)** 16	Add the two values.
4	**(b)** 35	Multiply the coefficient by the value of x.
5	**(c)** 14	Replace x by its value and calculate.
6	**(a)** 7	Replace y by its value and subtract.
7	**(b)** $k/4$	Translate the words into a letter-number expression.
8	**(d)** 2	Terms are separated by + or - signs.
9	**(a)** 12	The coefficient is the number multiplying the letter.
10	**(d)** $6p$ and p	Like terms have the same letter part.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Group the like terms (1); Sum $= 8x + 11$ (1).	1+1
12	Substitute: $6(4) + 2$ (1); $= 26$ (1).	1+1
13	Identify the operations (1); Expression: $20 - x$ (1).	1+1
14	Group the like terms (1); Sum $= 8x + 14$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Substitute: $6(3) + 5(2)$ (1); $= 18 + 10$ (1); $= 28$ (1).	1+1+1
16	All three terms are like terms (1); Coefficients: $10 + 10 - 4 = 16$ (1); Answer: $16x$ (1).	1+1+1
17	Cost $= 12p$ (1); At $p = 8$: 12×8 (1); Cost $=$ ₹96 (1).	1+1+1
	OR: Substitute: $7(4) + 2(6)$ (1); $= 28 + 12$ (1); $= 40$ (1).	1+1+1
18	All three terms are like terms (1); Coefficients: $8 + 7 - 6 = 9$ (1); Answer: $9x$ (1).	1+1+1
	OR: Cost $= 7p$ (1); At $p = 14$: 7×14 (1); Cost $=$ ₹98 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sister's age $= x - 6$ (1); Sum $= x + (x - 6) = 2x - 6$ (1); At $x = 25$: sister $= 19$ years (1); Sum $= 44$ years (2).	1+1+1+2
	OR: Perimeter $= 2(l + 7)$ (1); Area $= 7l$ (1); At $l = 11$: perimeter $= 36$ cm (1); Area $= 77$ cm^2 (2).	1+1+1+2
20	Sister's age $= x - 7$ (1); Sum $= x + (x - 7) = 2x - 7$ (1); At $x = 22$: sister $= 15$ years (1); Sum $= 37$ years (2).	1+1+1+2
	OR: Perimeter $= 2(l + 2)$ (1); Area $= 2l$ (1); At $l = 15$: perimeter $= 34$ cm (1); Area $= 30$ cm^2 (2).	1+1+1+2

Mistakes to avoid:

- Adding unlike terms: $4x + 3y = 7xy$ is wrong.
- Writing $3a + 2a = 5a^2$; the letter part does not change when like terms are added.
- Writing $3n$ as '3 plus n'; it means 3 times n.
- Not changing all signs when subtracting an expression in brackets: $(5x + 3) - (2x - 4) \neq 3x - 1$.
- Confusing 'n less than a' $(a - n)$ with $n - a$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.